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chapter

20 Carbohydrate Biosynthesis in Plants and Bacteria

1. **Segregation of Metabolism in Organelles** What are the advantages to the plant cell of having different organelles to carry out different reaction sequences that share intermediates?

   Answer Within organelles, reaction intermediates and enzymes can be maintained at different levels from those in the cytosol and in other organelles. For example, the ATP/ADP ratio is lower in mitochondria than in the cytosol because the role of adenine nucleotides in the mitochondrial matrix is to accept a phosphoryl group, whereas the role in the cytosol is to donate a phosphoryl group. Similarly, different NADH/NAD^+ and NADPH/$NADP^+$ ratios reflect the reductive (biosynthetic) functions of the cytosol and the oxidative (catabolic) functions of the mitochondrial matrix. By segregating reaction sequences that share intermediates, the cell can regulate catabolic and anabolic processes separately.

2. **Phases of Photosynthesis** When a suspension of green algae is illuminated in the absence of CO_2 and then incubated with $^{14}CO_2$ in the dark, $^{14}CO_2$ is converted to [^{14}C]glucose for a brief time. What is the significance of this observation with regard to the CO_2-assimilation process, and how is it related to the light reactions of photosynthesis? Why does the conversion of $^{14}CO_2$ to [^{14}C]glucose stop after a brief time?

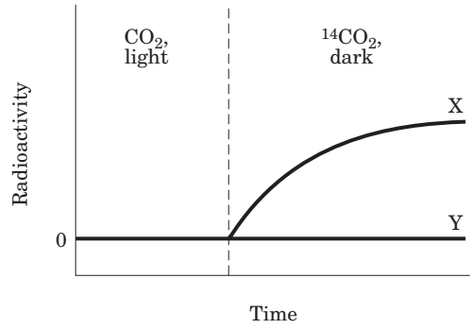
   Answer This observation suggests that photosynthesis occurs in two phases: (1) a light-dependent phase that generates ATP and NADPH, which are essential for CO_2 fixation, and (2) a light-independent (dark) phase, in which these energy-rich components are used for synthesis of glucose. In the absence of additional illumination, the supplies of NADPH and ATP become exhausted and CO_2 fixation ceases.

3. **Identification of Key Intermediates in CO_2 Assimilation** Calvin and his colleagues used the unicellular green alga *Chlorella* to study the carbon-assimilation reactions of photosynthesis. They incubated $^{14}CO_2$ with illuminated suspensions of algae and followed the time course of appearance of ^{14}C in two compounds, X and Y, under two sets of conditions. Suggest the identities of X and Y, based on your understanding of the Calvin cycle.
   (a) Illuminated *Chlorella* were grown with unlabeled CO_2, then the light was turned off and $^{14}CO_2$ was added (vertical dashed line in graph (a)). Under these conditions, X was the first compound to become labeled with ^{14}C; Y was unlabeled.
   (b) Illuminated *Chlorella* cells were grown with $^{14}CO_2$. Illumination was continued until all the $^{14}CO_2$ had disappeared (vertical dashed line in graph (b)). Under these conditions, X became labeled quickly but lost its radioactivity with time, whereas Y became more radioactive with time.

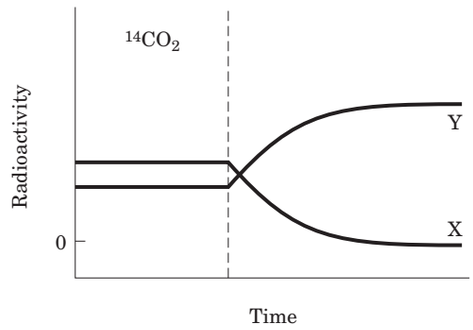
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(a)



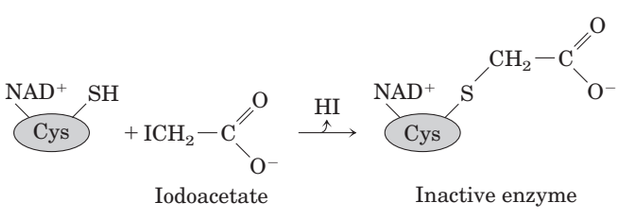
(b)

Answer Compound X is 3-phosphoglycerate and compound Y is ribulose 1,5-bisphosphate (see Fig. 20–4).

(a) Illumination of *Chlorella* in the presence of unlabeled CO_2 gives rise to steady-state levels of ribulose 1,5–bisphosphate, 3–phosphoglycerate, ATP, and NADPH. When the light is turned off, the production of ATP and NADPH ceases, but the Calvin cycle continues briefly until the residual ATP and NADPH are exhausted. Once this occurs, conversion of 3-phosphoglycerate (Stage 2 in Fig. 20–4) to hexoses, which depends on ATP and NADPH, is blocked. Thus, $^{14}CO_2$ added at the time the light is turned off is transformed primarily to 3-phosphoglycerate, but not to other intermediates of the cycle such as ribulose 1,5-bisphosphate.

(b) Illumination of *Chlorella* in the presence of $^{14}CO_2$ gives rise to steady-state levels of ^{14}C-labeled 3-phosphoglycerate and ribulose 1,5-bisphosphate. If the concentration of CO_2 is rapidly decreased, none is available for the ribulose 1,5-bisphosphate carboxylase reaction, and this constitutes a block at the fixation step (Stage 1 in Fig. 20–4). Because this experiment is carried out under conditions of constant illumination, the steps requiring ATP and NADPH are not blocked and all labeled 3-phosphoglycerate (X) can be converted to labeled ribulose 1,5-bisphosphate (Y). This results in a decrease in labeled X and a commensurate increase in labeled Y.

4. Regulation of the Calvin Cycle Iodoacetate reacts irreversibly with the free —SH groups of Cys residues in proteins.



Iodoacetate Inactive enzyme

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Predict which Calvin cycle enzyme(s) would be inhibited by iodoacetate, and explain why.

Answer Ribulose 5-phosphate kinase, fructose 1,6-bisphosphatase, sedoheptulose 1,7-bisphosphatase, and glyceraldehyde 3-phosphate dehydrogenase would be inhibited. All have mechanisms requiring activation by reduction of a critical disulfide bond to a pair of —SH groups. Iodoacetate reacts irreversibly with free —SH groups.

5. **Thioredoxin in Regulation of Calvin Cycle Enzymes** Motohashi and colleagues used thioredoxin as a hook to fish out from plant extracts the proteins that are activated by thioredoxin. To do this, they prepared a mutant thioredoxin in which one of the reactive Cys residues was replaced with a Ser. Explain why this modification was necessary for their experiments. [Motohashi, K., Kondoh, A., Stumpp, M.T., & Hisabori, T. (2001) Comprehensive survey of proteins targeted by chloroplast thioredoxin. *Proc. Natl. Acad. Sci. USA* **98,** 11,224-11,229.]

   Answer The several Calvin cycle enzymes that are regulated by the effect of light on ferredoxin reduction are activated when reduced ferredoxin passes electrons to thioredoxin, reducing a disulfide bond to two —SH groups. For thioredoxin to play its catalytic role in the disulfide exchange reaction that activates the Calvin cycle enzymes, it must have both of its reactive —SH groups. The mutant thioredoxin, in which one of the reactive Cys residues is converted to a Ser (which lacks an —SH group), cannot form an internal disulfide, as it has only one —SH group. When this mutant thioredoxin interacts with one of the enzymes regulated by disulfide reduction, it forms a disulfide bond with that enzyme, but the next step, the reduction of the pair of Cys residues in the enzyme, cannot occur because the thioredoxin cannot form a disulfide. Consequently, thioredoxin remains covalently attached to the enzyme, and when the thioredoxin is isolated, the enzyme comes along, "hooked" on thioredoxin.

6. **Comparison of the Reductive and Oxidative Pentose Phosphate Pathways** The *reductive* pentose phosphate pathway generates a number of intermediates identical to those of the *oxidative* pentose phosphate pathway (Chapter 14). What role does each pathway play in cells where it is active?

   Answer The *reductive* pentose phosphate pathway regenerates ribulose 1,5-bisphosphate from triose phosphates produced during photosynthesis, in a series of reactions involving sugars of three, four, five, six, and seven carbons and the enzymes transaldolase and transketolase. The oxidative pentose phosphate pathway plays a different metabolic role: it provides NADPH for reductive biosynthesis and pentose phosphates for nucleotide synthesis.

7. **Photorespiration and Mitochondrial Respiration** Compare the oxidative photosynthetic carbon cycle (C_2 cycle), also called *photorespiration,* with the *mitochondrial respiration* that drives ATP synthesis. Why are both processes referred to as respiration? Where in the cell do they occur, and under what circumstances? What is the path of electron flow in each?

   Answer Both processes are called "respiration" because both consume O_2 and produce CO_2. Mitochondrial respiration, or cellular respiration, occurs only in mitochondria, providing most of the ATP required by the cell. In the mitochondrion, electrons derived from oxidation of various fuels in the matrix are passed through a chain of carriers in the inner mitochondrial membrane to O_2 (see Fig. 19–16); the proton gradient thus generated provides the energy for ATP synthesis. Photorespiration takes place only in plants. It is the long series of reactions, involving enzymes in the chloroplasts and peroxisomes, in which the glycolate produced when rubisco "fixes" molecular oxygen is converted to phosphoglycerate. In this reaction series (see Fig. 20–21), one-fourth of the carbons in glycolate are released as CO_2, and O_2 is consumed in the reaction catalyzed by glycolic acid oxidase. This amount of O_2 consumption is added to that occurring in the rubisco reaction. Photorespiration is an expensive side reaction of photosynthesis, resulting from the lack of specificity of rubisco.

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8. Rubisco and the Composition of the Atmosphere N. E. Tolbert has argued that the dual specificity of rubisco for CO_2 and O_2 is not simply a leftover from evolution in a low-oxygen environment. He suggests that the relative activities of the carboxylase and oxygenase activities of rubisco actually have set, and now maintain, the ratio of CO_2 to O_2 in the earth's atmosphere. Discuss the pros and cons of this hypothesis, in molecular terms and in global terms. How does the existence of C_4 organisms bear on the hypothesis? [Tolbert, N.E. (1994) The role of photosynthesis and photorespiration in regulating atmospheric CO_2 and O_2. In *Regulation of Atmospheric CO_2 and O_2 by Photosynthetic Carbon Metabolism* (Tolbert, N.E. & Preiss, J., eds), pp. 8–33, Oxford University Press, New York.]

Answer This hypothesis assumes that the rubisco reaction is the most important chemical reaction in determining the composition of the earth's atmosphere. This may once have been true, but in the modern world, many other processes, such as the burning of fossil fuels and destruction of tropical forests (by burning and other means), probably have at least as large an impact on atmospheric composition. To the extent that C4 plants contribute to global CO_2 fixation, they also tend to counterbalance the effect of rubisco on the balance of CO_2 and O_2 in the atmosphere.

9. Role of Sedoheptulose 1,7-Bisphosphatase What effect on the cell and the organism might result from a defect in sedoheptulose 1,7-bisphosphatase in **(a)** a human hepatocyte and **(b)** the leaf cell of a green plant?

Answer

(a) In a human hepatocyte, NADPH produced by the pentose phosphate pathway—for which sedoheptulose 1,7-bisphosphatase activity is essential—is needed for the reductive steps in the synthesis of fatty acids, triacylglycerols, and sterols. Without this source of NADPH, all these processes would be inhibited. Cells would be unable to synthesize lipids and other reduced products.

(b) The lack of sedoheptulose 1,7-bisphosphatase would prevent the functioning of the Calvin cycle; the enzyme is essential in the series of reactions that convert triose phosphates to ribulose 1,5-bisphosphate. Without generation of ribulose 1,5-bisphosphate, the Calvin cycle is effectively blocked; there is no acceptor for CO_2 in the first step in that pathway. The leaf cell would therefore be incapable of fixing CO_2.

10. Pathway of CO_2 Assimilation in Maize If a maize (corn) plant is illuminated in the presence of $^{14}CO_2$, after about 1 second, more than 90% of all the radioactivity incorporated in the leaves is found at C-4 of malate, aspartate, and oxaloacetate. Only after 60 seconds does ^{14}C appear at C-1 of 3-phosphoglycerate. Explain.

Answer In maize, CO_2 is fixed by the C_4 pathway elucidated by Hatch and Slack. Phosphoenolpyruvate is rapidly carboxylated to oxaloacetate, some of which undergoes transamination to aspartate but most of which is reduced to malate in the mesophyll cells. Only after subsequent decarboxylation of labeled malate does $^{14}CO_2$ enter the Calvin cycle for conversion to glucose. The rate of entry into the cycle is limited by the rate of the rubisco-catalyzed reaction.

11. Identifying CAM Plants Given some $^{14}CO_2$ and all the tools typically present in a biochemistry research lab, how would you design a simple experiment to determine whether a plant was a typical C_4 plant or a CAM plant?

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Answer The distinguishing features of CAM metabolism are the initial fixation of CO_2 at night and the storage of this fixed CO_2 in the vacuole until the next morning, when photosynthesis begins again. One could therefore measure the amount of $^{14}CO_2$ fixed in leaves during an hour of darkness, and the amount fixed during an hour of bright illumination. The CAM plant will take up much more CO_2 at night than a typical C_4 plant. One could also measure the concentration of organic acids in the vacuoles by titrating an extract of leaves; the CAM plant will have (in darkness) a much higher level of titratable acidity due to the malic acid stored in the vacuoles.

12. Chemistry of Malic Enzyme: Variation on a Theme Malic enzyme, found in the bundle-sheath cells of C_4 plants, carries out a reaction that has a counterpart in the citric acid cycle. What is the analogous reaction? Explain your choice.

Answer Malic enzyme catalyzes oxidative decarboxylation of a hydroxycarboxylic acid in the C_4 pathway:

$$\underset{\text{Malate}}{{}^-OOC{-}CH(OH){-}CH_2{-}COO^-} + NADP^+ \longrightarrow \underset{\text{Pyruvate}}{{}^-OOC{-}CO{-}CH_3} + CO_2 + NADPH + H^+$$

which is analogous to the reaction catalyzed in the citric acid cycle by the enzyme isocitrate dehydrogenase:

$$\text{Isocitrate} + NAD^+ \longrightarrow \alpha\text{-ketoglutarate} + NADH + H^+$$

13. The Cost of Storing Glucose as Starch Write the sequence of steps and the net reaction required to calculate the cost, in ATP molecules, of converting a molecule of cytosolic glucose 6-phosphate to starch and back to glucose 6-phosphate. What fraction of the maximum number of ATP molecules available from complete catabolism of glucose 6-phosphate to CO_2 and H_2O does this cost represent?

Answer The reactions for formation of starch from glucose 6-phosphate are:

(1) Glucose 6-phosphate $\rightarrow$ glucose 1-phosphate

(2) ATP + glucose 1-phosphate $\rightarrow$ ADP-glucose + PP_i

(3) ADP-glucose + $\text{starch}_n \rightarrow$ ADP + starch_{n+1}

The reactions for breakdown of starch are:

(4) $\text{Starch}_{n+1} + P_i \rightarrow \text{starch}_n$ + glucose 1-phosphate

(5) Glucose 1-phosphate $\rightarrow$ glucose 6-phosphate

The sum of reactions (1) through (5) is

(6) ATP + $P_i \rightarrow PP_i$ + ADP

Because it takes one ATP to convert ADP to ATP, we need to add

(7) ADP + ATP $\rightarrow$ ATP + ADP

The net reaction is the sum of reactions (6) and (7):

(8) ATP + $P_i \rightarrow$ ADP + PP_i

So, 1 mol of ATP is expended to store 1 mol of glucose 6-phosphate as starch. A mole of glucose 6-phosphate, when fully oxidized via glycolysis and the citric acid cycle, yields 31 to 33 mol of ATP, so the cost of storage is 1/33, or about 3.3% of the energy available.

14. Inorganic Pyrophosphatase The enzyme inorganic pyrophosphatase contributes to making many biosynthetic reactions that generate inorganic pyrophosphate essentially irreversible in cells. By keeping the concentration of PP_i very low, the enzyme "pulls" these reactions in the direction of PP_i formation. The synthesis of ADP-glucose in chloroplasts is one reaction that is pulled in the forward direction by this mechanism. However, the synthesis of UDP-glucose in the plant cytosol, which produces PP_i, is readily reversible in vivo. How do you reconcile these two facts?

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Answer [PP_i] is high in the plant cell cytosol because the cytosol lacks inorganic pyrophosphatase, the enzyme that degrades PP_i to 2 P_i. Plants are unique in this regard. Animal cells have pyrophosphatase in their cytosol, and [PP_i] is therefore kept too low for PP_i to be a useful phosphoryl group donor.

15. Regulation of Starch and Sucrose Synthesis Sucrose synthesis occurs in the cytosol and starch synthesis in the chloroplast stroma, yet the two processes are intricately balanced. What factors shift the reactions in favor of **(a)** starch synthesis and **(b)** sucrose synthesis?

Answer

(a) Low levels of P_i in the cytosol and high levels of triose phosphate in the chloroplast favor formation of starch.

(b) High levels of triose phosphate in the cytosol favor formation of sucrose.

16. Regulation of Sucrose Synthesis In the regulation of sucrose synthesis from the triose phosphates produced during photosynthesis, 3-phosphoglycerate and P_i play critical roles (see Fig. 20–26). Explain why the concentrations of these two regulators reflect the rate of photosynthesis.

Answer 3-Phosphoglycerate is the primary product of photosynthesis; [P_i] rises when light-driven synthesis of ATP from ADP and P_i slows. The concentrations of these two metabolites thus provide clues to the energetic state of the leaf cell. When photosynthesis is occurring at a high rate, [P_i] drops and [3-phosphoglycerate] rises; in the dark, [P_i] rises and [3-phosphoglycerate] falls. The rate-limiting step in sucrose synthesis is the formation of ADP-glucose from glucose 1-phosphate and ATP, and this step is inhibited by P_i and activated by 3-phosphoglycerate (see Fig. 20–28). Sucrose synthesis therefore occurs at a high rate when photosynthesis is occurring, [P_i] is low, and [3-phosphoglycerate] is high.

17. Sucrose and Dental Caries The most prevalent infection in humans worldwide is dental caries, which stems from the colonization and destruction of tooth enamel by a variety of acidifying microorganisms. These organisms synthesize and live within a water-insoluble network of dextrans, called dental plaque, composed of ($\alpha 1 \rightarrow 6$)-linked polymers of glucose with many ($\alpha 1 \rightarrow 3$) branch points. Polymerization of dextran requires dietary sucrose, and the reaction is catalyzed by a bacterial enzyme, dextran-sucrose glucosyltransferase.

(a) Write the overall reaction for dextran polymerization.

(b) In addition to providing a substrate for the formation of dental plaque, how does dietary sucrose also provide oral bacteria with an abundant source of metabolic energy?

Answer

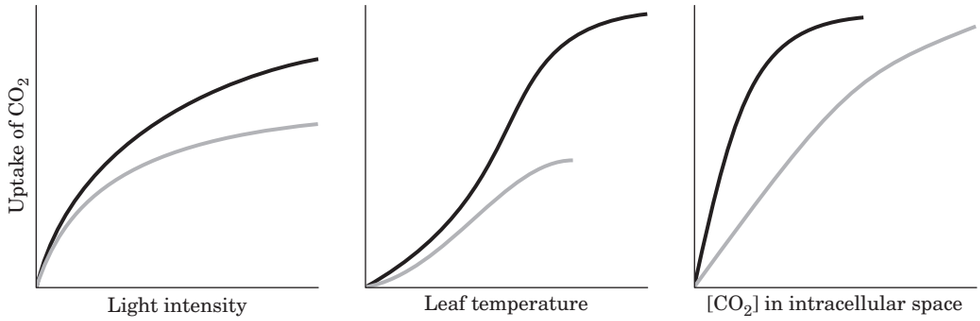
(a) $$\text{Sucrose} + \underset{\text{dextran}_n}{(\text{glucose})_n} \longrightarrow \underset{\text{dextran}_{n+1}}{(\text{glucose})_{n+1}} + \text{fructose}$$

(b) Fructose generated in the synthesis of dextran is readily taken up by the bacteria and metabolized to acidic compounds.

18. Differences between C_3 and C_4 Plants The plant genus *Atriplex* includes some C_3 and some C_4 species. From the data in the plots below (species 1, upper curve; species 2, lower curve), identify which is a C_3 plant and which is a C_4 plant. Justify your answer in molecular terms that account for the data in all three plots.

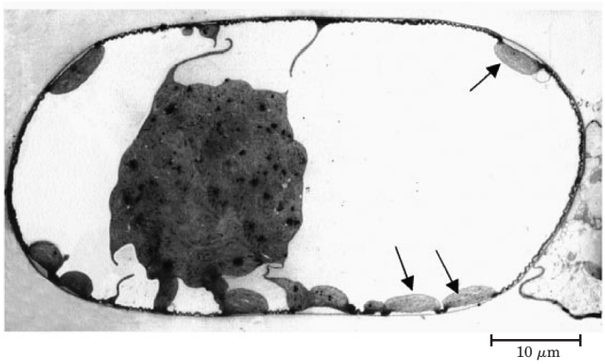
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Answer Species 1 is the C_4 variety; species 2, the C_3 variety. The "mistaken" use of O_2 instead of CO_2 by rubisco is greater at higher light intensities (higher photosynthetic rates) and at higher temperatures, so under these conditions the C_4 plant has the advantage. In the rightmost panel, the fixation of CO_2 is measured as a function of $[CO_2]$, and again the C_4 plant has the advantage. The lower K_m for CO_2 of PEP carboxylase (relative to rubisco) gives the C_4 plant the advantage at lower CO_2 concentrations.

19. C_4 Pathway in a Single Cell In typical C_4 plants, the initial capture of CO_2 occurs in one cell type, and the Calvin cycle reactions occur in another (see Fig. 20–23). Voznesenskaya and colleagues have described a plant, *Bienertia cycloptera*—which grows in salty depressions of semidesert in Central Asia—that shows the biochemical properties of a C_4 plant but unlike typical C_4 plants does not segregate the reactions of CO_2 fixation into two cell types. PEP carboxylase and rubisco are present in the same cell. However, the cells have two types of chloroplasts, which are localized differently, as shown in the micrograph. One type, relatively poor in grana (thylakoids), is confined to the periphery; the more typical chloroplasts are clustered in the center of the cell, separated from the peripheral chloroplasts by large vacuoles. Thin cytosolic bridges pass through the vacuoles, connecting the peripheral and central cytosol. [Voznesenskaya, E.V., Fraceschi, V.R., Kiirats, O., Artyusheva, E.G., Freitag, H., & Edwards, G.E. (2002) Proof of C_4 photosynthesis without Kranz anatomy in *Bienertia cycloptera* (Chenopodiaceae). *Plant J.* **31,** 649–662.]



In this plant, where would you expect to find **(a)** PEP carboxylase, **(b)** rubisco, and **(c)** starch granules? Explain your answers with a model for CO_2 fixation in these C_4 cells.

Answer In this organism, the initial capture of CO_2 and the final CO_2 fixation by rubisco take place not in different cell types but in two different regions of the same cell. The enzymes for initial fixation of CO_2 into oxaloacetate are at the periphery, where the concentrations of CO_2 and O_2 are relatively high; rubisco is in the cluster of chloroplasts at the center of the cell, where the O_2 concentration is low, limited by diffusion into the cell.

(a) Periphery of the cell.

(b) Chloroplasts at center of the cell.

(c) Chloroplasts at center of the cell.

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Data Analysis Problem

20. Rubisco of Bacterial Endosymbionts of Hydrothermal Vent Animals Undersea hydrothermal vents support remarkable ecosystems. At these extreme depths there is no light to support photosynthesis, yet thriving vent communities are found. Much of their primary productivity occurs through chemosynthesis carried out by bacterial symbionts that live in specialized organs (trophosomes) of certain vent animals.

Chemosynthesis in these bacteria involves a process that is virtually identical to photosynthesis. Carbon dioxide is fixed by rubisco and reduced to glucose, and the necessary ATP and NADPH are produced by electron-transfer processes similar to those of the light-dependent reactions of photosynthesis. The key difference is that in chemosynthesis, the energy driving electron transfer comes from a highly exergonic chemical reaction rather than from light. Different chemosynthetic bacteria use different reactions for this purpose. The bacteria found in hydrothermal vent animals typically use the oxidation of H_2S (abundant in the vent water) by O_2, producing elemental sulfur. These bacteria also use the conversion of H_2S to sulfur as a source of electrons for chemosynthetic CO_2 reduction.

(a) What is the overall reaction for chemosynthesis in these bacteria? You do not need to write a balanced equation; just give the starting materials and products.

(b) Ultimately, these endosymbiotic bacteria obtain their energy from sunlight. Explain how this occurs.

Robinson and colleagues (2003) explored the properties of rubisco from the bacterial endosymbiont of the giant tube worm *Riftia pachyptila*. Rubisco, from any source, catalyzes the reaction of either CO_2 (Fig. 20–7) or O_2 (Fig. 20–20) with ribulose 1,5-bisphosphate. In general, rubisco reacts more readily with CO_2 than O_2. The degree of selectivity (Ω) can be expressed as

$$\frac{V_{\text{carboxylation}}}{V_{\text{oxygenation}}} = \Omega\,\frac{[CO_2]}{[O_2]}$$

where V is the reaction velocity.

Robinson and coworkers measured the Ω value for the rubisco of the bacterial endosymbionts. They purified rubisco from tube-worm trophosomes, reacted it with mixtures of different ratios of O_2 and CO_2 in the presence of [1-^{3}H]ribulose 1,5-bisphosphate, and measured the ratio of [^{3}H]phosphoglycerate to [^{3}H]phosphoglycolate.

(c) The measured ratio of [^{3}H]phosphoglycerate to [^{3}H]phosphoglycolate is equal to the ratio $V_{\text{carboxylation}}/V_{\text{oxygenation}}$. Explain why.

(d) Why would [5-^{3}H]ribulose 1,5-bisphosphate not be a suitable substrate for this assay?

The Ω for the endosymbiont rubisco had a value of 8.6 ± 0.9.

(e) The atmospheric (molar) concentration of O_2 is 20% and that of CO_2 is about 380 parts per million. If the endosymbiont were to carry out chemosynthesis under these atmospheric conditions, what would be the value of $V_{\text{carboxylation}}/V_{\text{oxygenation}}$?

(f) Based on your answer to **(e),** would you expect Ω for the rubisco of a terrestrial plant to be higher than, the same as, or lower than 8.6? Explain your reasoning.

Two stable isotopes of carbon are commonly found in the environment: the more abundant ^{12}C and the rare ^{13}C. All rubisco enzymes catalyze the fixation of $^{12}CO_2$ faster than that of $^{13}CO_2$. As a result, the carbon in glucose is slightly enriched in ^{12}C compared with the isotopic composition of CO_2 in the environment. Several factors are involved in this "preferential" use of $^{12}CO_2$, but one factor is the fundamental physics of gases. The temperature of a gas is related to the kinetic energy of its molecules. Kinetic energy is given by $\frac{1}{2}mv^2$, where m is molecular mass and v is velocity. Thus, at the same temperature (same kinetic energy), the molecules of a lighter gas will be moving faster than those of a heavier gas.

(g) How could this contribute to rubisco's "preference" for $^{12}CO_2$ over $^{13}CO_2$?

Some of the first convincing evidence that the tube-worm hosts were obtaining their fixed carbon from the endosymbionts was that the $^{13}C/^{12}C$ ratio in the animals was much closer to that of the bacteria than that of nonvent marine animals.

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(h) Why is this more convincing evidence for a symbiotic relationship than earlier studies that simply showed the presence of rubisco in the bacteria found in trophosomes?

Answer

(a) By analogy to the oxygenic photosynthesis carried out by plants ($H_2O + CO_2 \rightarrow$ glucose $+ O_2$), the reaction would be $H_2S + O_2 + CO_2 \rightarrow$ glucose $+ H_2O + S$. This is the sum of the reduction of CO_2 by H_2S ($H_2S + CO_2 \rightarrow$ glucose $+ S$) and the energy input ($H_2S + O_2 \rightarrow S + H_2O$).

(b) The H_2S and CO_2 are produced chemically in deep-sea sediments, but the O_2, like the vast majority of O_2 on Earth, is produced by photosynthesis, which is driven by light energy.

(c) In the assay used by Robinson et al., ^3H labels the C-1 of ribulose 1,5-bisphosphate, so reaction with CO_2 yields one molecule of [^{3}H]3-phosphoglycerate and one molecule of unlabeled 3-phosphoglycerate; reaction with O_2 produces one molecule of [^{3}H]2-phosphoglycolate and one molecule of unlabeled 3-phosphoglycerate. Thus the ratio of [^{3}H]3-phosphoglycerate to [^{3}H]2-phosphoglycolate equals the ratio of carboxylation to oxygenation.

(d) If the ^3H labeled C-5, *both* oxygenation and carboxylation would yield [^{3}H]3-phosphoglycerate and it would be impossible to distinguish which reaction had produced the labeled product; the reaction could not be used to measure Ω.

(e) Substituting $\frac{[CO_2]}{[O_2]} = \frac{0.00038}{0.2} = 0.0019$ into $\frac{V_{\text{carboxylation}}}{V_{\text{oxygenation}}} = \Omega \frac{[CO_2]}{[O_2]}$ gives

$$\frac{V_{\text{carboxylation}}}{V_{\text{oxygenation}}} = (8.6)(0.0019) = 0.016$$

Therefore, the rate of oxygenation would be roughly 60 times the rate of carboxylation!

(f) If terrestrial plants had $\Omega = 8.6$, carboxylation would occur at a much lower rate than oxygenation. This would be extremely inefficient, so one would expect the rubisco of terrestrial plants to have an Ω substantially higher than 8.6. In fact, Ω values for land plants vary between 10 and 250. Even with these values, the expected rate of the oxygenation reaction is still very high.

(g) The rubisco reaction occurs with CO_2 as a gas. At the same temperature, $^{13}CO_2$ molecules diffuse more slowly than the lighter $^{12}CO_2$ molecules, and thus $^{13}CO_2$ will enter the active site (and become incorporated into substrate) more slowly than $^{12}CO_2$.

(h) For the relationship to be truly symbiotic, the tube worms must be getting a substantial amount of their carbon from the bacteria. The presence of rubisco in the endosymbionts simply shows that they are capable of chemosynthesis, not that they are supplying the host with a significant fraction of its carbon. On the other hand, showing that the $^{13}C : {}^{12}C$ ratio in the host is more similar to that in the endosymbiont than that in other marine animals strongly suggests that the tube worms are getting the majority of their carbon from the bacteria.

Reference

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